

Ahmed Fathy Gwely

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CAREER OBJECTIVE

Passionate about leveraging Artificial Intelligence, Computer Vision, and IoT to develop innovative solutions for real-world challenges, with a focus on deep learning and smart automation.

EDUCATION

Electrical Engineering, Mansoura University

10/2021 – 06/2026

COURSES / TRAINING

Intro to Programming Nano Degree

[Certificate from Udacity](#)

Data Analysis with Python

IBM

Data Visualization with Python

IBM

Applied Data Science Capstone

IBM

Health Safety and Environmental HSE

UCCD Mansoura University

Training Of Trainer (TOT)

Be a Professional Trainer

AI Overview

Huawei

Introduction to Artificial Intelligence and its Applications

zewail city

Supervised Machine Learning: Regression and Classification

Stanford & DeepLearning.AI

Computer vision

Opencv , Mediapipe, YOLOv8 , CNN

AI / IOT Engineer Samsung Innovation Campus

- AI Engineer Program – (2024)
- IoT Engineer Program – (2025)

Introducing Generative AI with AWS – Udacity

Covered Amazon Bedrock, SageMaker, and CodeWhisperer with applications in real-world generative AI use cases.

Building LLM Applications With Prompt Engineering

NVIDIA

Oracle Cloud Infrastructure 2025 Certified Data Science

Professional

Data Science Professional

Oracle Cloud Infrastructure 2025 Certified Generative AI

Professional

Generative AI Professional

Building RAG Agents with LLMs

NVIDIA

EXPERIENCE

Intern at Lotus Power Company, Computer Vision Engineer

- Developed **computer vision** and **machine learning** models for **IoT-based** smart farm management devices. Collaborated with embedded systems, **power electronics**, and software teams to implement real-time farm monitoring solutions with reliable on-device processing. Assisted in research, documentation, and patent/publication contributions for innovative agricultural technol

12/2025 – Present
Cairo, Egypt & Vancouver,
BC, Canada (Remote)

AI & IoT Engineer – Hyper SCADA (Egyptian Joint-Stock Company), Freelance

Designed Computer Vision & IoT projects in company system in water sector to automate processes and reduce human intervention, improving efficiency and cutting costs.

04/2024

Information Technology Institute (ITI), Generative AI Engineer Trainee

NVIDIA DLI Summer Training Program: Generative AI at the Information Technology Institute (ITI).

09/2025 – 10/2025

Neurobotics, AI & Computer Vision Instructor

AI Instructor and Technical Management

01/2025 – Present

Intern Samsung Innovation Campus IOT Engineering, IOT Engineer

IOT & Embedded Linux Engineer

08/2025

Smart City

Intern Samsung Innovation Campus AI Engineering, AI Engineer

AI Engineer with Top 2 Leaderboard Positions in Machine Learning and Deep Learning Competitions

07/2024 – 10/2024

Giza

Zewail City Computing Society (ZCCS) Researcher, Undergraduate Researcher

Artificial Intelligence Undergraduate Researcher

11/2024

Zewail City

IEEE Mansoura Computer Society

IEEE Mansoura Computer Society

Deep learning Supervisor in Data Science Committee, Mentor , Instructor

12/2024

Ambassador for Technology Innovation and Entrepreneurship Center (TIEC), Ambassador

TIEC at Ministry of Communication & Technology

Mansoura University Student Activities, Head of AI Committee / AI Computer Vision Instructor / Public Speaker

Momentum MU :

Head & CO-Founder Of AI Committee , Vice Training and public Speaker

2021 – 2025

Mansoura

IEEE Mansoura Computer Society

Deep learning Supervisor in Data Science Committee, Mentor , Instructor

CAT Reloaded

Deep Learning Supervisor at Data Science Committee for Advanced old Engineers Student

NextCoder Scholarship – Ministry of Communications and Information Technology (MCIT), Egypt ,
AI / Data Science Scholarship (Started in 1st year of secondary school)

06/2019 – 08/2020

PROJECTS

Developed a computer vision model for water station automation. ,

11/2025

Freelance Computer Vision Engineer at HyperScada

Built an end-to-end AI system using YOLO and MQTT for real-time water station automation, including custom data collection, model training, and live video inference.

Smart Poultry Farm Monitoring – YOLOv8 + IoT sensors for real-time chicken detection, tracking, and farm environment monitoring., for Lotus power company

Generate Anime Faces with GANs , Pytorch

Implement a Neural Style Transfer Application ,

using VGG19 and PyTorch with a Gradio interface, exploring deep learning for creative AI while emphasizing ethical considerations around artists' rights.

Developed a Deep Dream model ,

using pretrained CNNs (VGG19, Inception, ResNet) to generate dream-like visualizations and enhance image features.

Neural Network Digital Recognition from Scratch with NumPy only. ,

Neural Network from Scratch with NumPy only

Breast Cancer Detection – CNN + GANs model for tumor detection and classification from medical images. ,

Real Time Garbage Detection Computer Vision, Using Tuned Yolo8n

Pose Estimation – Human action recognition (e.g., drinking, gestures) using Mediapipe. ,

Smart Emotion-Detecting Sunglasses – Raspberry Pi wearable using CV for real-time emotion detection with audio feedback to assist blind users.

Object Detection Model – YOLOv8-based vehicle and object detection system (cars, trucks, buses). ,

Temple Run Gesture Control – Computer Vision + Mediapipe system for controlling Temple Run with hand gestures. ,

Sign Language Detection , computer vision / DL Sign Language Detection

Visual Sign-Controlled Electrical System – Gesture-based control of lights and devices using CV + IoT. ,

face detection - hand Detection - color detection, Computer Vision Projects & AI

Micro Mouse Robot – Autonomous maze-solving robot with AI path planning and IoT-based remote monitoring.

Brain Tumor Segmentation – Deep learning for automatic MRI-based tumor detection and segmentation. ,

House sales in king Country, Pandas - Python - Numpy - Matplotlib

Basic CV Projects – Implemented face, hand, and color detection using OpenCV.

SKILLS

Computer Vision Application | Image Processing | ML / DL Modeling and Deployment | Data Science | Deep Learning |
Electrical Hardware Deployment | Problem Solving | Data Visualization | Power BI | Leadership | Team Work |
Public Speaking (Training) | Network Architecture | GANs (Generative Adversarial Networks) | Embedded Linux | IOT |
Presenter and Instructor | MQTT | Knowledge of Wireless Communication Protocols | Micro Controllers (Raspberry Pi/ESP / etc...)

COMPETITIONS

Top 2 – Samsung SIC in Machine & Deep Learning Competitions .

10/2024

Winner 2nd Place – Industry 4.0 Transformation Marathon – Awarded 20,000 EGP.

04/2024

IEEE VICTORIS 2.0 – Micro Mouse Maze Solver. ,

08/2023

Smart Environment Hackathon – AI-powered sustainability solutions. ,

BOOKS

- Deep Learning With Python